

LAB NO 3

DATA STRUCTURES AND ALGORITHMS

OBJECTIVE: To understand the complexities of the recursive functions and a way to reduce these complexities

LAB TASKS

1. Write a program which takes an integer value (k) as input and prints the sequence of numbers from k to 0 in descending order.

Input 01:

```
public static void descendingOrder(int n) {
    if (n < 0) {
        return; // Base case: stops when n is less than 0
    }
    else{
        System.out.print(n + " "); // current num ko print karega pehle
        descendingOrder(n - 1); // Recursive case call hoga.
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Please enter a number: ");
    int k = sc.nextInt();
    System.out.println("Printing my agw in decending_order :");

    descendingOrder(k);
}
```

Output 01:

```
Output - LAB_NO_3_DSA (run) #4

run:
Please enter a number: 19
Printing my agw in decending_order :
19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 BUILD SUCCESSFUL (total time: 5 seconds)
```

2. Write a program to reverse your full name using Recursion.

Input 02:

```
// Write a program to reverse your full name using Recursion.

public static void reverse(String name){

    if(name.isEmpty() ){           // base case
        return ;
    }
    else{
        System.out.println(name.charAt(name.length()-1) + " ");
        reverse(name.substring(0, name.length()-1));    // calls recursive
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Please enter your name : ");
    String name = sc.nextLine();
    System.out.println("Printing your name in reversed_order :");
    reverse(name);
}
}
```

Output 02:

```
Output - LAB_NO_3_DSA (run)

run:
Please enter your name : WAQAR
Printing your name in reversed_order :
R
A
Q
A
W
BUILD SUCCESSFUL (total time: 5 seconds)
|
```

3. Write a program to calculate the sum of numbers from 1 to N using recursion. N should be user input

Input 03:

```
public static int calc(int n) {  
    if (n < 0) {  
        return 0;  
    }  
    else{  
        return n + calc(n-1);  
    }  
}  
  
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Please enter a number: ");  
    int n = sc.nextInt();  
    System.out.println("Calculating the sum:");  
    int result = calc(n);  
    System.out.println("The sum from " + n + " to 0 is: " + result);  
    calc(n);  
}
```

Output 03:

```
Output - LAB_NO_3_DSA (run)  
run:  
Please enter a number: 5  
Calculating the sum:  
The sum from 1 to 5 is: 15  
BUILD SUCCESSFUL (total time: 3 seconds)  
|
```

4. Write a recursive program to calculate the sum of elements in an array.

Input 04:

```
// Write a recursive program to calculate the sum of elements in an array.

public static int sum_arr(int arr[] , int n) {

    if (n < 0) {
        return 0;
    }
    else{
        return arr[n] + sum_arr(arr, n - 1);
    }
}

public static void main(String[] args) {

    int arr[] = {10,20,30,40,50};
    System.out.println("Calculating the sum:");

    int result = sum_arr(arr, arr.length-1);

    System.out.println("The sum of array is: " + result);
}
```

Output 04:

```
Output - LAB_NO_3_DSA (run)

run:
Calculating the sum:
The sum of array is: 150
BUILD SUCCESSFUL (total time: 0 seconds)
```

5. Write a recursive program to calculate the factorial of a given integer n

Input 05:

```
if(n==0) {  
    return 1;  
}  
else{  
    return n * calc_fact(n-1);  
}  
}  
  
public static void main(String[] args) {  
  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Please enter a number: ");  
    int n = sc.nextInt();  
    System.out.println("Calculating the Factorial:");  
    int result = calc_fact(n);  
    System.out.println("The fcac of " + n + " is: " + result);  
    calc_fact(n);  
}
```

Output 05:

```
Output - LAB_NO_3_DSA (run)  
  
run:  
Please enter a number: 5  
Calculating the Factorial:  
The fcac of 5 is: 120  
BUILD SUCCESSFUL (total time: 2 seconds)
```

6. Write a program to count the digits of a given number using recursion.

Input 06:

```
// Write a program to count the digits of a given number using recursion.

public static int count(String n ){

    if(n.isEmpty()){

        return 0;

    }

    else{

        return 1 + count(n.substring(1));

    }

}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Please enter a number: ");

    String n = sc.nextLine();

    System.out.println("Counting the numbers:");

    int result = count(n);

    System.out.println("The counted numbers are: " + result);

    count(n);

}

}
```

Output 06:

```
Output - LAB_NO_3_DSA (run)

run:
Please enter a number: 03104919293
Counting the numbers:
The counted numbers are: 11
BUILD SUCCESSFUL (total time: 21 seconds)
```

HOME TASKS

1. Write a java program to find the N-th term in the Fibonacci series using Memoization.

Input 01:

```
static int arr[] = new int[1000]; // if a number is calculated then it is to be stored inside any array
// and not to calculate it again we // have used the logic in else to check is it already calculated or not

public static int fib(int n){
    if (n == 0 || n == 1)
        return n;

    else{
        if(arr[n] != 0){ // checks if the fibonacci number is already calculated
            return arr[n];
        }
        else{
            arr[n] = fib(n-1) + fib(n-2); // if not then calculates the number
            return arr[n];
        }
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Please enter a number: ");
    int n = sc.nextInt();
    int result = fib(n);
    System.out.println("the result is: " + result);
}
```

Output 01:

```
Output - LAB_NO_3_DSA (run)

run:
Please enter a number: 10
the result is: 55
BUILD SUCCESSFUL (total time: 1 second)
```

2. Write a java program to check whether a given string is a palindrome or not. A palindrome is a string that reads the same forwards and backwards. Print "YES" if the string is a palindrome, otherwise print "NO"

Input 02:

```
public static boolean isPalindrome(String str) {
    String str1 = str.toLowerCase(); // converts to lower case

    // Reverse the string
    String reversedStr = new StringBuilder(str1).reverse().toString();

    // Checks if the original string is same as the reversed string
    return str1.equals(reversedStr);
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Please enter a string: ");
    String input = sc.nextLine();

    if (isPalindrome(input)) {
        System.out.println("The string is a palindrome.");
    }
    else {
        System.out.println("The string is not a palindrome.");
    }
}
```

Output 02:

```
Output - LAB_NO_3_DSA (run)

run:
Please enter a string: LOL
The string is a palindrome.
BUILD SUCCESSFUL (total time: 11 seconds)
```


3. Write a recursive program to find the greatest common divisor (GCD) of two numbers using Euclid's algorithm

Input 03:

```
public static int gcd(int a, int b) {  
    if (b == 0) {  
        return a;  
    }  
  
    return gcd(b, a % b);    // Recursive case:  
}  
  
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
  
    System.out.print("Enter the first number: ");  
    int a = sc.nextInt();  
    System.out.print("Enter the second number: ");  
    int b = sc.nextInt();  
  
    int result = gcd(a, b);  
    System.out.println("The GCD of " + a + " and " + b + " is: " + result);  
}  
}
```

Output 03:

```
Output - LAB_NO_3_DSA (run) #3  
  
run:  
Enter the first number: 81  
Enter the second number: 63  
The GCD of 81 and 63 is: 9  
BUILD SUCCESSFUL (total time: 6 seconds)
```